

SQL Expert Practice Questions 101-120

Very hard SQLite challenges with focused hints



EXPERT CONCEPTS

- Recursive CTEs and organisational hierarchies
- Median, percentile and NTILE analysis
- Gaps-and-islands and consecutive-month streaks
- Pareto analysis, complete calendars and moving averages
- Relational division, set equality and RFM segmentation

Question-and-hint edition | Solutions are intentionally excluded for independent practice

How to approach these questions

These challenges are designed to be solved in stages. Build and test each CTE separately before combining the final query.

1	Define the grain Decide whether each row represents an employee, customer, product, category or month.
2	Build intermediate results Aggregate before ranking and create one logical CTE per transformation.
3	Test boundaries Check ties, NULLs, empty sets, missing months and threshold-crossing rows.
4	Explain the method Be ready to describe why the chosen window frame, recursion or set comparison works.

EXPERT LEVEL 1 Advanced Employee Analytics

Department benchmarks, medians, hierarchies, percentiles and salary segmentation.

Q101

Department salary band

Find employees whose salary is above their department average but below the highest salary in their department.

Hint: Calculate the department average and maximum without losing employee-level rows.

AVG OVER / MAX OVER

Q102

Median salary by department

Find the median salary in each department.

Hint: Number ordered salaries, count department rows, then average the middle one or two values.

ROW_NUMBER / MEDIAN LOGIC

Q103

Direct and indirect reports

Find every manager's direct and indirect reports, including hierarchy level.

Hint: Start with direct reports, then recursively follow each report's own subordinates.

RECURSIVE CTE

Q104

Total reporting-tree size

Find the total number of direct and indirect reports for each manager.

Hint: Reuse the recursive hierarchy and count distinct descendants per manager.

RECURSIVE CTE / COUNT

Q105

Department salary percentile

Find the salary percentile of every employee within their department.

Hint: Partition by department, order by salary, and convert PERCENT_RANK() to a percentage.

PERCENT_RANK

Q106

Department salary groups

Divide employees in each department into three salary groups.

Hint: Use NTILE(3) after partitioning by department and ordering salary from highest to lowest.

NTILE

EXPERT LEVEL 2 Advanced Customer Behaviour

Sequential behaviour, order gaps, relational division, streaks and favourite products.

Q107

Continuously increasing customer spend

Find customers whose order amount increased with every successive order.

Hint: Use LAG() to compare each order with the previous one, then reject any non-increasing sequence.

LAG / CONDITIONAL AGGREGATION

Q108

Longest gap between orders

Find each customer's longest gap in days between consecutive orders.

Hint: Use LAG() for the prior date, julianday() for the gap, and RANK() for longest ties.

LAG / JULIANDAY / RANK

Q109**Customers covering every category**

Find customers who purchased every product category available.

Hint: Compare each customer's distinct purchased-category count with the total available category count.

RELATIONAL DIVISION

Q110**Customers buying every Electronics product**

Find customers who purchased every product in the Electronics category.

Hint: Search for an Electronics product missing from the customer's orders, then keep customers with no missing product.

DOUBLE NOT EXISTS

Q111**Longest consecutive-month streak**

Find each customer's longest streak of consecutive months containing at least one order.

Hint: Convert months to a continuous number, subtract ROW_NUMBER(), then group equal differences into streaks.

GAPS AND ISLANDS

Q112**Favourite product with ties**

Find each customer's most frequently purchased product and return all tied products.

Hint: Count customer-product purchases first, then rank counts within each customer.

RANK / MULTI-STAGE CTE

EXPERT LEVEL 3 Revenue and Time-Series Analytics

Category comparisons, Pareto thresholds, peaks, complete calendars and moving averages.

Q113**Product versus category average**

Compare each product's revenue with its category's average product revenue.

Hint: Aggregate product revenue first, then apply AVG() OVER category to the aggregated result.

WINDOW AGGREGATE

Q114**Category-level Pareto products**

Find the products required to reach the first 80% of revenue within each category.

Hint: Order product revenue within category and compare the prior cumulative total with the 80% threshold.

PARETO / RUNNING TOTAL

Q115**Local monthly sales peaks**

Find months where total sales were higher than both the previous and following months.

Hint: Aggregate by year-month, then compare each total using both LAG() and LEAD().

LAG / LEAD

Q116**Complete monthly calendar**

Generate every month between the earliest and latest order and include months with zero sales.

Hint: Build months recursively, aggregate actual sales, then LEFT JOIN and replace missing values with zero.

RECURSIVE CALENDAR

Q117

Three-month moving average

Calculate a three-month moving average of monthly sales, including missing months as zero.

Hint: Create a complete month series, then use ROWS BETWEEN 2 PRECEDING AND CURRENT ROW.

MOVING WINDOW

EXPERT LEVEL 4 Advanced Business Intelligence

Cumulative contribution, set equality and RFM customer segmentation.

Q118

Smallest group reaching 50% of revenue

Find the smallest group of highest-spending customers needed to reach at least 50% of total revenue.

Hint: Calculate running revenue in descending spend order and include the customer who crosses the boundary.

CUMULATIVE CONTRIBUTION

Q119

Customers with identical product sets

Find pairs of customers who purchased exactly the same distinct set of products.

Hint: Generate customer pairs, then use EXCEPT in both directions to confirm neither set has extra products.

CROSS JOIN / EXCEPT

Q120

RFM customer segmentation

Build an RFM-style report using recency, frequency and monetary value, then assign customer segments.

Hint: Use the latest order date as the reference, score active customers with NTILE(), and classify by combined scores.

RFM / NTILE / CASE

Final review checklist

- I tested each CTE independently.
- My ranking function handles ties as intended.
- My recursive CTE has a valid anchor and termination condition.
- Missing months and NULL values are handled explicitly.
- I can explain the query in clear business language.